MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 05** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 20**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

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Q1. From a point on the ground which is 30 m away from the foot of a tower, the angle of elevation of the top of the tower is 30°. The height of the tower is: [1]

(A) $10\sqrt{3}$ m (B) $20\sqrt{3}$ m (C) $30\sqrt{3}$ m (D) 10 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $h = 30\tan 30° = \dfrac{30}{\sqrt{3}} = \boxed{10\sqrt{3}\text{ m}}$.

Q2. Two vertical poles of heights 6 m and 11 m stand on flat ground. If the distance between their feet is 12 m, the distance between their tops is: [1]

(A) 11 m (B) 12 m (C) 13 m (D) 14 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Justification: Vertical height difference $= 11 - 6 = 5$ m, horizontal separation $= 12$ m. By Pythagoras theorem, distance $= \sqrt{12^2 + 5^2} = \sqrt{169} = \boxed{13\text{ m}}$.

Q3. If the altitude of the sun changes from 30° to 60°, the length of the shadow of a 15 m high pole: [1]

(A) increases by $10\sqrt{3}$ m (B) decreases by $10\sqrt{3}$ m (C) decreases by $5\sqrt{3}$ m (D) remains same

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Justification: $s_{30} = 15\sqrt{3}$ m, $s_{60} = \dfrac{15}{\sqrt{3}} = 5\sqrt{3}$ m. Change $= 15\sqrt{3} - 5\sqrt{3} = 10\sqrt{3}$ m shorter, so it $\boxed{\text{decreases by } 10\sqrt{3}\text{ m}}$.

Q4. If the height of a tower and the distance of the point of observation from its foot both are increased by 10%, then the angle of elevation of its top: [1]

(A) increases (B) decreases (C) remains unchanged (D) is halved

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Justification: $\tan\theta' = \dfrac{1.10\,h}{1.10\,d} = \dfrac{h}{d} = \tan\theta \implies \theta' = \theta$, so the angle $\boxed{\text{remains unchanged}}$.

Q5. **Assertion (A):** If the height of a vertical tower is doubled, the angle of elevation of its top from a fixed point on the ground does not double.

Reason (R): The value of $\tan\theta = \dfrac{h}{d}$ doubles when h doubles, but $\tan(2\theta) \neq 2\tan\theta$ for acute angles. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $\tan\theta_{\text{new}} = 2\tan\theta$, whereas $\tan(2\theta) = \dfrac{2\tan\theta}{1-\tan^2\theta} > 2\tan\theta$ for $0 < \theta < 45°$. Both (A) and (R) are true and (R) explains (A).

SECTION B: Short Answer Questions

MATHS MANTHAN | (11 Marks)

Q6. If a tower 30 m high casts a shadow $10\sqrt{3}$ m long on the ground, calculate the angle of elevation of the sun. [2]

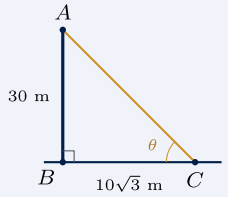
▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let tower $AB = 30$ m cast shadow $BC = 10\sqrt{3}$ m, and sun's angle of elevation be $\angle ACB = \theta$. [0.5 Mark]

In right $\triangle ABC$: $\tan\theta = \dfrac{AB}{BC} = \dfrac{30}{10\sqrt{3}} = \dfrac{3}{\sqrt{3}} = \sqrt{3}$. [1 Mark]

$$\because\ \tan 60° = \sqrt{3} \implies \theta = \boxed{60°}$$

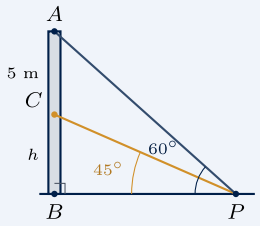
Final sun elevation: [0.5 Mark]



Q7. A vertical flagstaff of height 5 m stands on top of a tower. From a point on the ground, the angles of elevation of the top and bottom of the flagstaff are 60° and 45° respectively. Find the height of the tower. [Take $\sqrt{3} = 1.732$] **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let tower be $BC = h$ m and flagstaff be $AC = 5$ m, observed from ground point P with $\angle CPB = 45°$ and $\angle APB = 60°$.



[0.5 Mark]

In right $\triangle CBP$: $\tan 45° = \dfrac{h}{BP} \implies BP = h.$ [1 Mark]

In right $\triangle ABP$: $\tan 60° = \dfrac{h+5}{BP} \implies \sqrt{3} = \dfrac{h+5}{h}.$ [0.5 Mark]

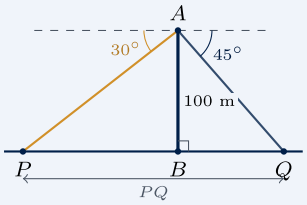
$$h(\sqrt{3}-1) = 5 \implies h = \frac{5}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{5(\sqrt{3}+1)}{2} = 2.5(\sqrt{3}+1) \text{ m.}$$ [0.5 Mark]

With $\sqrt{3} = 1.732$: $h = 2.5(2.732) = \boxed{6.83 \text{ m}}$. [0.5 Mark]

Q8. An observer on top of a cliff 100 m high observes two boats in the sea on opposite sides of the cliff. If the angles of depression of the boats are 30° and 45°, find the distance between the two boats. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let cliff $AB = 100$ m, with boat P (left, depression 30° $\implies$ $\angle APB = 30°$) and boat Q (right, depression 45° $\implies$ $\angle AQB = 45°$). [0.5 Mark]



In right $\triangle ABP$: $PB = 100 \cot 30° = 100\sqrt{3}$ m. [1 Mark]

In right $\triangle ABQ$: $BQ = 100 \cot 45° = 100$ m. [1 Mark]

Distance between the boats $= PQ = PB + BQ = 100\sqrt{3} + 100 = 100(\sqrt{3}+1)$ m.

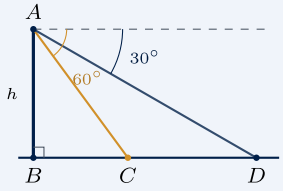
$$PQ = 100(1.732 + 1) = 100 \times 2.732 = \boxed{100(\sqrt{3}+1) \text{ m} \approx 273.2 \text{ m}}$$

Final distance: [0.5 Mark]

Q9. A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30°, which is approaching the foot of the tower with a uniform speed. Six minutes later, the angle of depression of the car is found to be 60°. Find the time taken by the car to reach the foot of the tower from this point. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let tower $AB = h$, initial car position D ($\angle ADB = 30°$), position after 6 min be C ($\angle ACB = 60°$), uniform speed v. [0.5 Mark]



In $\triangle ABC$: $BC = \dfrac{h}{\sqrt{3}}$; in $\triangle ABD$: $BD = h\sqrt{3}$. Thus $DC = BD - BC = h\sqrt{3} - \dfrac{h}{\sqrt{3}} = \dfrac{2h}{\sqrt{3}}$ in 6 min. [1.5 Marks]

Notice $DC = 2\,BC$. Since speed is uniform and DC takes 6 minutes, the remaining half distance CB from the 60° point C to foot B takes:

$$\text{Time from } C \text{ to foot } B = \frac{6}{2} = \boxed{3 \text{ minutes}}$$

Final time from 60° point: [1 Mark]

Q10. CASE STUDY: Coast Guard Airborne Search and Rescue Operation

An Indian Coast Guard helicopter H hovers stationary at altitude $HP = 1200$ m directly above checkpoint P. It spots a life raft R at an angle of depression of 30° and a rescue speed-boat B approaching R along the same radial line at an angle of depression of 60°.

(i) Find the horizontal distance between the checkpoint P and the rescue speed-boat B. [1]

(ii) Find the direct line-of-sight distance from the helicopter H to the life raft R. [1]

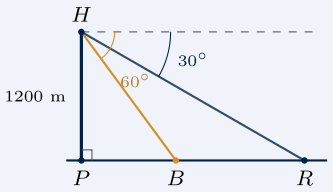
(iii) Calculate the distance between the speed-boat and the life raft (i.e., length BR). If the boat travels at a speed of 20 m/s, how many seconds will it take to reach the raft? [Take $\sqrt{3} = 1.732$] [2]

OR

If the helicopter ascends vertically by 300 m to reach an altitude of 1500 m, what will be the horizontal distance to an object observed at an angle of depression of 45°? [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) In right $\triangle HPB$ ($\angle HBP = 60^\circ, HP = 1200$ m):



$$\tan 60^\circ = \frac{1200}{PB} \implies PB = \frac{1200}{\sqrt{3}} = \boxed{400\sqrt{3} \text{ m} \approx 692.8 \text{ m}}$$

Subpart (i): [1 Mark]

(ii) In right $\triangle HPR$ ($\angle HRP = 30^\circ$):

$$\sin 30^\circ = \frac{HP}{HR} \implies \frac{1}{2} = \frac{1200}{HR} \implies HR = \boxed{2400 \text{ m}}$$

Subpart (ii): [1 Mark]

(iii) Main Choice: $PR = 1200 \cot 30^\circ = 1200\sqrt{3}$ m. Hence $BR = PR - PB = 1200\sqrt{3} - 400\sqrt{3} = 800\sqrt{3} = 800(1.732) = \boxed{1385.6 \text{ m}}$. [1 Mark]

Time at 20 m/s $= \dfrac{1385.6}{20} = \boxed{69.28 \text{ seconds}}$. [1 Mark]

(iii) OR Choice: New altitude $= 1200 + 300 = 1500$ m. At depression 45°:

$$\tan 45^\circ = \frac{1500}{d} \implies 1 = \frac{1500}{d} \implies d = \boxed{1500 \text{ m}}$$

New horizontal distance: [2 Marks]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★